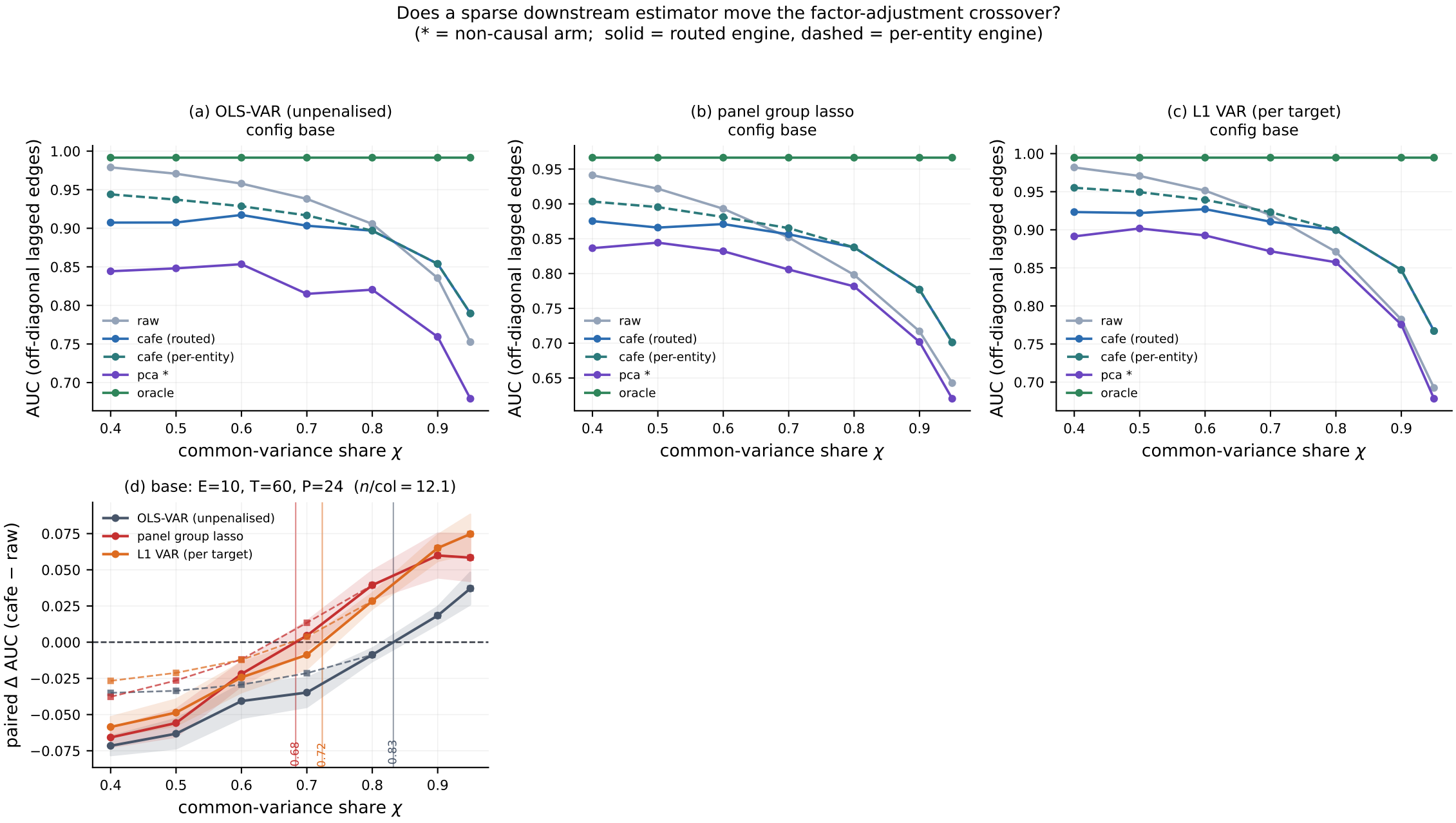
Does a sparse downstream estimator move the factor-adjustment crossover?
(* = non-causal arm; solid = routed engine, dashed = per-entity engine)
(a) OLS-VAR (unpenalised)
config base
(b) panel group lasso
config base
(c) L1 VAR (per target)
config base
AUC (off-diagonal lagged edges)
common-variance share χ
raw
cafe (routed)
cafe (per-entity)
pca *
oracle
(d) base: E=10, T=60, P=24 (n/col = 12.1)
paired Δ AUC (cafe − raw)
OLS-VAR (unpenalised)
panel group lasso
L1 VAR (per target)
0.68
0.72
0.83